Strict global PF_4 for the Riemann kernel and a continuous Fourier separator

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Abstract

Let Φ be the classical Riemann, or de Bruijn–Newman, kernel. We prove that every translation minor of Φ of order at most four is strictly positive at strictly ordered nodes, and a direct origin-confluent calculation gives a negative order-five minor. Thus the exact global Polya-frequency order is four. For the Riemann-kernel positive theorem, the computer-assisted input is confined to directed one-variable inequalities proving global positivity of the log-curvature quantities q, F_2 and of the fully confluent invariant $\mathcal{C}_4$. The order-five origin certificate uses exact rational arithmetic. A quotient-Wronskian argument reduces arbitrary fourth-order minors to $\partial_\xi \Psi < 0$. In the increasing curvature coordinate, its numerator is

$$\delta\Lambda \int_p^w W(t) \frac{\mathcal{C}_4(t)}{Q(t)^6 \kappa(t)^2} \, \mathrm{d}t.$$

Here W is the difference of two explicitly normalized triangular CDFs. Its strict positivity follows from a one-crossing density ratio. The structural transport reductions are displayed in the text and appendices; the compact covers, theta-tail paddings, and separator coefficient reconstruction remain explicitly identified certificate boundaries. We then construct an explicit positive even Schwartz kernel which is itself strictly PF_4 but whose entire Fourier transform has nonreal zeros; the latter sign uses one directed 192-bit discriminant enclosure. Thus continuous PF_3 and PF_4 membership alone do not force Fourier real-rootedness. The results are finite-order and have no implication for the Riemann Hypothesis.

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Contents

1 Introduction

A function $f : \mathbb{R} \to \mathbb{R}$ is PF_r when every translation-kernel minor

$$\det[f(x_i - y_j)]_{i,j=1}^{k}$$

is nonnegative for $1 \leq k \leq r$ and increasing node tuples. We call the property strict when the determinant is positive whenever both tuples are strictly increasing. This paper determines the exact finite Polya-frequency order of the Riemann kernel; see [1] for modern background and Schoenberg's characterization. Related finite-order and Riemann-kernel work is placed in Section 2.

Theorem 1.1 (Strict global PF_4)**.** *For every $1 \leq k \leq 4$ and strictly increasing real tuples $x_1 < \cdots < x_k$ and $y_1 < \cdots < y_k$,*

$$\det[\Phi(x_i - y_j)]_{i,j=1}^{k} > 0.$$

A parity factorization of the fully confluent derivative matrix at the origin gives a negative order-five limit. Divided differences then transfer that sign to equally spaced translation minors at sufficiently small positive spacing; see Theorem 11.2. Hence Theorem 1.1 has the following exact consequence.

Corollary 1.2. *The exact global Polya-frequency order of Φ is four.*

The exact-order classification and the Fourier question are distinct. At the proved order-four level, the following example shows that finite-order membership alone does not force Fourier real-rootedness, even within the positive even Schwartz class.

Theorem 1.3 (Continuous PF_4 Fourier separator)**.** *There is an explicit positive even Schwartz function f_* which is strict* PF_4*, while its entire Fourier transform has nonreal zeros. Consequently continuous* PF_3 *and continuous* PF_4 *membership are each insufficient to force Fourier real-rootedness.*

The proof has two certified one-variable inputs and one exact transport mechanism. First, global inequalities $q > 0$ and $F_2 > 0$ imply strict PF_3 and positivity of a three-point functional Λ. Second, the fully confluent order-four invariant $\mathcal{C}_4$ is globally positive. Exact quotient identities reduce strict PF_4 to the monotonicity of a scalar three-point function Ψ. An increasing curvature coordinate then turns the derivative of Ψ into a positive crossing-kernel integral of $\mathcal{C}_4$.

The distinction between a human derivation and a computer certificate is maintained throughout. Directed interval arithmetic proves the compact and tail inequalities stated in Section 4; the structural transport algebra is written in the paper and independently replayed by repository verifiers. The exact separator coefficient list is printed in Appendix E. The short order-five certificate uses only exact rational arithmetic and elementary series bounds. The classification concerns finite-order total positivity only. It neither proves nor disproves the Riemann Hypothesis. Theorem 1.3, proved in Section 12, supplies an explicit order-four counterexample rather than a finite-order sufficiency threshold.

2 Relation to earlier work

We use *finite-order* Polya-frequency terminology in the sense reviewed by Belton, Guillot, Khare, and Putinar [1]. Khare's characterization of measurable TN_p functions for $p \geq 3$ gives a broader finite-order context for translation-kernel criteria [2]; the present proof instead derives a kernel-specific strict order-four criterion through quotient Wronskians and a curvature transport identity.

For the same normalization of the Riemann kernel, Dimitrov and Xu relate Wronskians of Fourier and Laplace transforms to correlation kernels and real-zero criteria [3]. Csordas and Varga obtain strict concavity and moment inequalities for the Riemann kernel in connection with the Riemann Hypothesis [4]. Those works concern adjacent Wronskian, moment, and Fourier questions. Our theorem here is the finite translation-minor statement in Theorem 1.1; no comprehensive priority claim beyond that precise proved statement is made.

3 The kernel and smooth even continuation

Put

$$\vartheta(x) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 x}, \qquad \Theta(t) = \vartheta(e^{2t}), \qquad H(t) = e^{t/2}\Theta(t).$$

The theta transformation $\vartheta(x) = x^{-1/2}\vartheta(1/x)$ gives

$$\Theta(-t) = e^t\Theta(t), \qquad H(-t) = H(t).$$

The series and all its derivatives converge locally uniformly, so H is real analytic and even. Define

$$\Phi(t) = \frac{1}{2}\left(H''(t) - \frac{1}{4}H(t)\right).$$

Termwise differentiation gives, for every real t,

$$\Phi(t) = \sum_{n\geq 1}\left(4\pi^2 n^4 e^{9t/2} - 6\pi n^2 e^{5t/2}\right) e^{-\pi n^2 e^{2t}}. \tag{3.1}$$

Thus the positive-side formula often written with $|t|$ is the restriction of a real-analytic even function. In particular, smoothness at the origin is not inferred from numerical coverage.

For $t \geq 0$, the coefficient of the n-th exponential can be factored as

$$2\pi n^2 e^{5t/2}\left(2\pi n^2 e^{2t} - 3\right) > 0,$$

because $2\pi > 3$. Hence every term in (3.1) is positive and $\Phi(t) > 0$ on $[0, \infty)$. Evenness gives $\Phi(t) > 0$ on all of $\mathbb{R}$, so the following logarithm is globally defined.

Write

$$\ell = \log \Phi, \qquad s = \ell', \qquad q = -s' = -\ell''.$$

For $a < b$, define

$$A(a, b) = s(a) - s(b), \qquad M(a, b) = \frac{q(b) - q(a)}{A(a, b)}. \tag{3.2}$$

When $q > 0$, s is strictly decreasing and $A(a, b) > 0$.

4 Certified one-variable inputs

Define

$$F_1 = qq'' - (q')^2, \qquad F_2 = q^3 - F_1.$$

For the logarithmic cumulants $c_j = \ell^{(j)}$, set

$$\begin{aligned}
&m_0 = 1, \quad m_1 = 0, \quad m_2 = c_2, \quad m_3 = c_3,\\
&m_4 = c_4 + 3c_2^2, \quad m_5 = c_5 + 10c_2c_3,\\
&m_6 = c_6 + 15c_2c_4 + 10c_3^2 + 15c_2^3,
\end{aligned}$$

and

$$\mathcal{C}_4(t) = \det[m_{i+j-2}(t)]_{i,j=1}^4. \tag{4.1}$$

Lemma 4.1 (Certified input)**.** *For the kernel* Φ*,*

$$q(t) > 0, \qquad F_2(t) > 0, \qquad \mathcal{C}_4(t) > 0 \quad (t \in \mathbb{R}).$$

The compact interval $[0, 1]$ is covered with directed Arb balls at 192-bit precision. The retained lower bounds are

$$q \geq 18.7268, \qquad F_2 \geq 3889.2, \qquad \mathcal{C}_4 \geq 2.817 \times 10^7.$$

The first certificate uses 7731 adaptive second-order Taylor cells and the second 8050. Evenness covers $[-1, 1]$.

For $t \geq 1$, put $x = \pi e^{2t}$. The cumulants have directed enclosures

$$c_j = -2^j x + \varepsilon_j, \qquad |\varepsilon_j| \leq \frac{E_j}{x}, \qquad 2 \leq j \leq 6,$$

where

$$(E_2, E_3, E_4, E_5, E_6) = (19.8, 176, 2082, 30770, 545900).$$

They imply $q, F_2 > 0$ on the tail. Substitution in the determinant (4.1) gives

$$\frac{\mathcal{C}_4(t)}{x^6} > 44392 \qquad (x \geq \pi e^2).$$

The derivation of the E_j, the exact lower polynomial, and its monotonic normalized margin are recorded in Appendix B. These are the only directed numerical inputs to the main positive proof.

5 Order three and the positive slope functional

For $\xi < m < r$, define

$$\Lambda(\xi; m, r) = A(\xi, r) + M(\xi, m) - M(m, r). \tag{5.1}$$

We give the full estimate used to prove its positivity.

Set $f_0 = q'/q$. Because $s' = -q$,

$$M(a,b) = \frac{\int_a^b q'(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t} = \frac{\int_a^b f_0(t)q(t)\,\mathrm{d}t}{\int_a^b q(t)\,\mathrm{d}t};$$

thus $M(a,b)$ is a q-weighted mean of f_0. Choose a point $u \in [\xi, m]$ where f_0 is minimal on the left interval and a point $v \in [m, r]$ where it is maximal on the right. Then

$$M(m,r) - M(\xi, m) \leq f_0(v) - f_0(u) \leq \int_\xi^r \max(f_0'(t), 0)\,\mathrm{d}t.$$

Since

$$f_0' = \frac{qq'' - (q')^2}{q^2} = \frac{F_1}{q^2}$$

and $A(\xi, r) = \int_\xi^r q(t)\,\mathrm{d}t$, equation (5.1) yields

$$\Lambda(\xi; m, r) \geq \int_\xi^r \left(q - \frac{\max(F_1, 0)}{q^2}\right)\mathrm{d}t \tag{5.2}$$

$$= \int_\xi^r \frac{q^3 - \max(F_1, 0)}{q^2}\,\mathrm{d}t = \int_\xi^r \frac{\min(q^3, F_2)}{q^2}\,\mathrm{d}t > 0. \tag{5.3}$$

The last inequality follows from Lemma 4.1 and $q > 0$.

The same quantity is the logarithmic slope derivative in the normalized order-three determinant. Hence (5.3), together with strict log-concavity, proves strict global PF_3. We rederive the needed Wronskian form in the next section because its signs also control order four.

6 Quotient-Wronskian reduction

Fix $y_1 < y_2 < y_3 < y_4$, put $u_j(t) = \Phi(t - y_j)$, and write $p_j = t - y_j$, so $p_4 < p_3 < p_2 < p_1$. Define

$$v_j = (u_j/u_1)', \qquad w_j = (v_j/v_2)'.$$

Successive column division and differentiation give

$$W(u_1, u_2, u_3) = u_1^3 W(v_2, v_3), \tag{6.1}$$

$$W(u_1, u_2, u_3, u_4) = u_1^4 v_2^3 w_3^2 \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{w_4}{w_3}\right). \tag{6.2}$$

Let $\mathcal{T}$ denote simultaneous translation of all point arguments. From (3.2),

$$\mathcal{T} A(a,b) = q(b) - q(a), \qquad \mathcal{T} \log A(a,b) = M(a,b). \tag{6.3}$$

This sign is load-bearing. Since

$$\frac{v_j}{v_2} = \frac{u_j}{u_2}\frac{A(p_j,p_1)}{A(p_2,p_1)},$$

direct differentiation gives

$$g_j := \mathcal{T} \log(v_j/v_2) = A(p_j,p_2) + M(p_j,p_1) - M(p_2,p_1). \tag{6.4}$$

The exact weighted-mean split

$$A(p_j,p_1)M(p_j,p_1) = A(p_j,p_2)M(p_j,p_2) + A(p_2,p_1)M(p_2,p_1)$$

turns (6.4) into

$$g_j = \frac{A(p_j,p_2)}{A(p_j,p_1)}\Lambda(p_j;p_2,p_1) > 0. \tag{6.5}$$

Thus $w_j = (v_j/v_2)g_j > 0$, and (6.1) has the strict order-three sign.

Define

$$\Psi(\xi;m,r) = \Lambda(\xi;m,r) + \mathcal{T}\log\Lambda(\xi;m,r).$$

Because $w_j = (v_j/v_2)g_j$,

$$\mathcal{T}\log(w_4/w_3) = (g_4 + \mathcal{T}\log g_4) - (g_3 + \mathcal{T}\log g_3). \tag{6.6}$$

Equation (6.5) and (6.3) give

$$\mathcal{T}\log g_j = M(p_j,p_2) - M(p_j,p_1) + \mathcal{T}\log\Lambda_j,$$

where $\Lambda_j = \Lambda(p_j;p_2,p_1)$. A second use of the same mean split gives

$$g_j + M(p_j,p_2) - M(p_j,p_1) = \Lambda_j - A(p_2,p_1).$$

The common last term cancels in (6.6), yielding the exact identity

$$\mathcal{T}\log(w_4/w_3) = \Psi(p_4;p_2,p_1) - \Psi(p_3;p_2,p_1). \tag{6.7}$$

Proposition 6.1 (Three-point equivalence)**.** *Under the strict lower-order positivity already proved, global* PF_4 *is equivalent to*

$$\partial_\xi\Psi(\xi;m,r) \le 0 \qquad (\xi < m < r).$$

Strict inequality implies strict global PF_4*.*

Proof. If $\partial_\xi \Psi \le 0$, then $p_4 < p_3$ makes the right side of (6.7) nonnegative. All prefactors in (6.2) are positive. The Wronskian is therefore nonnegative, and is positive in the strict case.

For increasing row nodes $t_1 < \cdots < t_k$, divide columns by $u_1(t_i)$ and take consecutive row differences from the bottom upward. The fundamental theorem of calculus and determinant multilinearity give

$$\det[U_j(t_i)]_{i,j=1}^{k} = \int_{t_1}^{t_2} \cdots \int_{t_{k-1}}^{t_k} \det[U'_{j+1}(\tau_i)]_{i,j=1}^{k-1} \, \mathrm{d}\tau_1 \cdots \mathrm{d}\tau_{k-1},$$

where $U_1 = 1$. Repeating the quotient reduction converts the Wronskian sign to every collocation minor; positivity of the integrand on a product of positive-length intervals gives the strict conclusion.

Conversely, assume global PF_4. Let all four row nodes coalesce, divide by their positive Vandermonde, and pass to the limit. The resulting translate Wronskian is nonnegative. Equations (6.2) and (6.7) imply $\Psi(p_4; p_2, p_1) \ge \Psi(p_3; p_2, p_1)$ whenever $p_4 < p_3$. Thus $\Psi(\cdot; m, r)$ is nonincreasing, which is equivalent to the stated differential inequality because it is smooth. □

7 Curvature coordinate and sign bridge

Since $q > 0$,

$$y(t) = -s(t), \qquad Q(y(t)) = q(t)$$

defines a global strictly increasing coordinate on its image, with $\mathrm{d}y/\mathrm{d}t = q(t) = Q(y) > 0$. Primes in this and the next three sections mean y-derivatives. Put

$$\rho = 1 - Q'' = \frac{F_2}{q^3} > 0, \qquad \kappa = 2 - Q'' = 1 + \rho > 1.$$

For $\xi < m < r$, write

$$p = y(\xi), \quad z = y(m), \quad w = y(r), \quad L = z - p, \quad R = w - z.$$

Because $A(a, b) = y(b) - y(a)$ and $M(a, b) = Q[y(a), y(b)]$,

$$\Lambda = (w - p) + Q[p, z] - Q[z, w]. \tag{7.1}$$

Two integrations of $\kappa = 2 - Q''$ give

$$\Lambda = \frac{1}{L} \int_p^z (y - p)\kappa(y) \, \mathrm{d}y + \frac{1}{R} \int_z^w (w - y)\kappa(y) \, \mathrm{d}y, \tag{7.2}$$

$$\delta := -\partial_p \Lambda = \frac{1}{L^2} \int_p^z (z - y)\kappa(y) \, \mathrm{d}y > 0. \tag{7.3}$$

Simultaneous translation becomes

$$\mathcal{T} = Q(p)\partial_p + Q(z)\partial_z + Q(w)\partial_w,$$

and $\partial_\xi = Q(p)\partial_p$. Thus $\Lambda_\xi = -Q(p)\delta$.

Define

$$\mathcal{N} = \delta\Lambda^2 + Q'(p)\delta\Lambda + \Lambda\mathcal{T}\delta - \delta\mathcal{T}\Lambda. \tag{7.4}$$

Differentiating $\Psi = \Lambda + \mathcal{T} \log \Lambda$, commuting simultaneous translation with ∂_ξ, and collecting over Λ^2 gives

$$\partial_\xi \Psi = -\frac{Q(p)}{\Lambda^2} \mathcal{N}. \tag{7.5}$$

Equivalently,

$$\frac{\mathcal{N}}{\delta\Lambda} = K := \Lambda + Q'(p) + \mathcal{T}\log\delta - \mathcal{T}\log\Lambda. \tag{7.6}$$

It remains to prove $K > 0$.

8 The confluent invariant in curvature form

Define

$$D = 3(\kappa - 1) - \{Q(\log\kappa)'\}'. \tag{8.1}$$

Lemma 8.1 (Confluent-to-curvature identity)**.** *The determinant* (4.1) *satisfies*

$$\mathcal{C}_4 = Q^6\kappa^2 D. \tag{8.2}$$

Consequently $D > 0$ *on* $\mathbb{R}$.

Proof. The change of variable gives $\mathrm{d}/\mathrm{d}t = Q\,\mathrm{d}/\mathrm{d}y$ and $c_2 = -Q$. Recursively applying this operator produces the five cumulants listed in Appendix A. Substitution into the Schur-complement form of (4.1) factors out Q^6. The remaining factor is

$$\kappa^2\left(3(\kappa-1) - \{Q(\log\kappa)'\}'\right),$$

as shown by the common expanded numerator in Appendix A. This proves (8.2) without a numerical assumption. Lemma 4.1, $Q > 0$, and $\kappa > 0$ then imply $D > 0$. □

The determinant definition is primary: it is the doubly confluent fourth-order translation minor after division by the positive row and column Vandermonde factors and by Φ^4. The explicit thirteen-term expansion is included in Appendix A for independent checking of the certificate input.

9 Exact transport identity

Normalize the triangular weights in (7.2) and (7.3) as probability measures:

$$\mathrm{d}\mu(y) = \frac{(z-y)\kappa(y)}{L^2\delta}\mathbf{1}_{[p,z]}(y)\,\mathrm{d}y, \tag{9.1}$$

$$\mathrm{d}\nu(y) = \frac{(y-p)\kappa(y)}{L\Lambda}\mathbf{1}_{[p,z]}(y)\,\mathrm{d}y + \frac{(w-y)\kappa(y)}{R\Lambda}\mathbf{1}_{[z,w]}(y)\,\mathrm{d}y. \tag{9.2}$$

Set

$$A_0(y) = 3(y - Q'(y)) - Q(y)\frac{\kappa'(y)}{\kappa(y)}.$$

Then $A_0' = D$.

Lemma 9.1 (Transport cancellation)**.** *With* K *as in* (7.6)*,*

$$K = \mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0]. \tag{9.3}$$

Proof. Two elementary primitives carry the complete cancellation:

$$\kappa A_0 = H', \qquad H = 3y^2 - 3Q - 3yQ' + (Q')^2 + QQ'', \tag{9.4}$$

$$H = J', \qquad J = y^3 - 3yQ + QQ'. \tag{9.5}$$

Both identities follow by one differentiation.

Let $I_L = J(z) - J(p)$ and $I_R = J(w) - J(z)$. Integrating the linear weights in (9.1)–(9.2) by parts once gives

$$\mathbb{E}_\nu[A_0] - \mathbb{E}_\mu[A_0] = \frac{-I_L/L + I_R/R}{\Lambda} + \frac{LH(p) - I_L}{L^2\delta}. \tag{9.6}$$

On the other hand, differentiation of the triangular integrals along $\mathcal{T}$ gives

$$\mathcal{T}\Lambda = U(z,w) - U(p,z), \quad \mathcal{T}\delta = \frac{U(p,z) - (Q(z) - Q(p))\delta - Q(p)\kappa(p)}{L},$$

where

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d) - Q(c))^2}{(d-c)^2}. \tag{9.7}$$

Substitution in (7.6) yields

$$K = \Lambda + Q'(p) - Q[p,z] + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)\kappa(p)}{L\delta}. \tag{9.8}$$

The endpoint identity equating (9.6) and (9.8), with every substitution exposed, is Appendix C. It is an identity in the endpoint positions and the values of Q, Q', Q''; no bound or polynomial model is used. This proves (9.3). □

10 The positive crossing kernel

Let F_μ, F_ν be the CDFs of (9.1)–(9.2), and put

$$W(y) = F_\mu(y) - F_\nu(y).$$

Lemma 10.1 (Strict stochastic ordering)**.** *One has $W(y) > 0$ for $p < y < w$, while $W(p) = W(w) = 0$.*

Proof. On $[z,w]$, $F_\mu = 1$, so $W = 1 - F_\nu > 0$ before w. On (p,z), the ratio of the two positive densities is

$$R_0(y) = \frac{\mathrm{d}\mu/\,\mathrm{d}y}{\mathrm{d}\nu/\,\mathrm{d}y} = \frac{\Lambda(z-y)}{L\delta(y-p)}. \tag{10.1}$$

Writing $C = \Lambda/(L\delta) > 0$, direct differentiation gives

$$R_0'(y) = -\frac{C(z-p)}{(y-p)^2} < 0. \tag{10.2}$$

Thus R_0 decreases strictly from $+\infty$ to 0. Hence $W' = \mu' - \nu'$ changes sign exactly once, from positive to negative. Since $W(p) = 0$ and

$$W(z) = 1 - F_\nu(z) = \nu((z,w]) > 0,$$

the function cannot return to zero on $(p,z]$. The right interval was already handled. □

Integration by parts for CDFs, using $A_0' = D$ and the common total mass, turns Lemma 9.1 into

$$K = \int_p^w W(y)D(y)\,\mathrm{d}y. \tag{10.3}$$

Combining (7.6), (8.2), and (10.3) gives the central identity

$$\mathcal{N} = \delta\Lambda \int_p^w W(y)\frac{\mathcal{C}_4(y)}{Q(y)^6\kappa(y)^2}\,\mathrm{d}y > 0. \tag{10.4}$$

Every factor in the interior is strictly positive.

11 Completion and exact order

Equation (7.5) and (10.4) imply

$$\partial_\xi \Psi(\xi; m, r) < 0 \qquad (\xi < m < r).$$

Proposition 6.1 therefore proves Theorem 1.1.

We now prove directly that the classification stops at order four. Put

$$H_k = \det[\Phi^{(i+j)}(0)]_{i,j=0}^{k-1}, \qquad s_j = \Phi^{(j)}(0).$$

Since Φ is even, simultaneous parity permutations of the rows and columns give

$$H_4 = AB, \qquad H_5 = CB,$$

where

$$A = s_0 s_4 - s_2^2, \qquad B = s_2 s_6 - s_4^2,$$

and

$$C = \det \begin{pmatrix} s_0 & s_2 & s_4 \\ s_2 & s_4 & s_6 \\ s_4 & s_6 & s_8 \end{pmatrix}.$$

Lemma 11.1. *For the Riemann kernel,*

$$445 < A < 446, \qquad 189223 < B < 189228, \qquad -674000000 < C < -614000000.$$

In particular, $H_4 > 0$ and $H_5 < 0$.

Proof. For $x = \pi n^2 e^{2u}$, the n-th summand of $\Phi(u)$ is $e^{u/2-x} P_0(x)$, where $P_0(x) = 2x(2x-3)$. If its j-th derivative is $e^{u/2-x} P_j(x)$, then

$$P_{j+1} = 2xP_j' + (1/2 - 2x)P_j.$$

Thus

$$s_j = \sum_{n \geq 1} e^{-\pi n^2} P_j(\pi n^2).$$

The first three modes are enclosed by rational alternating-series bounds. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ supplies rational bounds for π, and argument division followed by the degree-29 and degree-30 alternating Taylor sums supplies rational bounds for each exponential.

For the omitted modes, let S_j be the sum of the absolute coefficients of P_j. Since $\deg P_j = j+2$, $3 < \pi < 22/7$, and $e^3 > \sum_{r=0}^{8} 3^r/r! > 20$,

$$\sum_{n \geq 4} |P_j(\pi n^2)| e^{-\pi n^2} \leq \frac{S_j (22/7)^{j+2} 4^{2j+4} 20^{-16}}{1 - (5/4)^{20} 20^{-9}} \qquad (j = 0, 2, 4, 6, 8).$$

Exact rational interval arithmetic then gives the stated bounds. The full endpoints and the standard-library verifier are included in the reproducibility archive. $\square$

Theorem 11.2. *The Riemann kernel is not* PF_5.

Proof. For a C^{2k-2} function f, repeated row and column divided differences give, for increasing fixed tuples a, b,

$$\lim_{h\downarrow 0} \frac{\det[f(z+h(a_i-b_j))]_{i,j=0}^{k-1}}{h^{k(k-1)}V(a)V(b)} = \frac{(-1)^{k(k-1)/2}}{\prod_{r=0}^{k-1}(r!)^2} \det[f^{(i+j)}(z)]_{i,j=0}^{k-1}.$$

Take $k = 5$, $z = 0$, and $a_i = b_i = i$. The sign factor is one, both Vandermonde factors are positive, and Lemma 11.1 makes the limit strictly negative. Hence $\det[\Phi((i-j)h)]_{i,j=0}^{4} < 0$ for every sufficiently small positive h. These are translation minors at strictly increasing nodes. $\square$

Theorem 1.1 and Theorem 11.2 prove Corollary 1.2.

The proof uses no multidimensional interval cover of the nonconfluent configuration space. Earlier positive-tail Peano densities, collision cones, Hermite boxes, and escape charts remain independent certified results, but they are not premises here. Their role was superseded when the transport numerator was recognized as the difference of expectations in (9.3).

Remark 11.3 (Riemann Hypothesis boundary). The theorem determines the kernel's exact finite Polya-frequency order and makes no claim about the zeros of the Riemann Ξ-function. In particular, the failure of PF_5 neither proves nor disproves the Riemann Hypothesis. The next section proves, by an explicit strict-PF_4 separator, that no Fourier real-rootedness conclusion follows from membership through order four alone.

12 An explicit continuous PF_4 separator

We now prove Theorem 1.3. Put

$$c = \frac{1}{64}, \qquad B(w) = \sum_{k=-3}^{3} b_k w^k, \qquad (b_{-3}, \ldots, b_3) = (2, 10, 23, 30, 23, 10, 2),$$

and define

$$f_*(x) = e^{-cx^2} B(e^{2cx}) = \sum_{k=-3}^{3} b_k e^{ck^2} e^{-c(x-k)^2}. \tag{12.1}$$

The Gaussian-mixture expression proves positivity and Schwartz decay; palindromy of (b_k) proves evenness.

12.1 Strict PF_3

For fixed x, give $K \in \{-3, \ldots, 3\}$ probability proportional to $b_k e^{2ckx}$, and write κ_j for its cumulants. Differentiation of the finite log-partition function gives

$$q = 2c - (2c)^2\kappa_2, \qquad q' = -(2c)^3\kappa_3, \qquad q'' = -(2c)^4\kappa_4. \tag{12.2}$$

If $\mu = \mathbb{E}K$, then $0 \le \mathbb{E}[(K+3)(3-K)] = 9 - \kappa_2 - \mu^2$. Thus $0 \le \kappa_2 \le 9$, and hence

$$q \ge 2c(1 - 18c) = 2c\frac{23}{32} > 0. \tag{12.3}$$

Moreover,

$$\begin{aligned} F_2 &= q^3 - \{qq'' - (q')^2\} \\ &= q^3 + q(2c)^4\kappa_4 + (2c)^6\kappa_3^2. \end{aligned} \tag{12.4}$$

The identity $\kappa_4 = \mathbb{E}(K - \mathbb{E}K)^4 - 3\kappa_2^2$ gives $\kappa_4 \geq -243$. Dropping the final nonnegative term in (12.4) gives

$$F_2 \geq q\{q^2 - 3888c^4\}.$$

At $c = 1/64$, the two coefficients after division by c^2 are

$$q^2/c^2 \geq \frac{529}{256}, \qquad 3888c^2 = \frac{243}{256}.$$

Thus $F_2 \geq qc^2(143/128) > 0$ globally. The weighted-mean criterion of Section 5 proves strict PF_3.

12.2 Exact order-four density

Let $\epsilon = 2c = 1/32$, $t = e^{\epsilon x}$, and

$$P(t) = t^3B(t) = 2 + 10t + 23t^2 + 30t^3 + 23t^4 + 10t^5 + 2t^6.$$

For $E = t\frac{\mathrm{d}}{\mathrm{d}t}$, define

$$N_1 = tP', \qquad N_{j+1} = t(N_j'P - jN_jP'). \tag{12.5}$$

The quotient rule gives $E^j \log P = N_j/P^j$. Therefore, for $\ell_j = (\log f_*)^{(j)}$,

$$\ell_2 = \frac{-\epsilon P^2 + \epsilon^2 N_2}{P^2}, \qquad \ell_j = \frac{\epsilon^j N_j}{P^j} \quad (3 \leq j \leq 6). \tag{12.6}$$

Insert these jets into the central-moment determinant (4.1). Every monomial has derivative weight twelve, so the common denominator is P^{12}:

$$\mathcal{C}_4[f_*](x) = \frac{H(t)}{P(t)^{12}}. \tag{12.7}$$

The exact expansion printed in Appendix E proves that H is palindromic of degree 72 and that all 73 coefficients are strictly positive. Hence $\mathcal{C}_4[f_*] > 0$ on $\mathbb{R}$.

The established inequalities $q > 0$ and $F_2 > 0$ also give

$$\kappa = 2 - Q'' = \frac{q^3 + F_2}{q^3} > 0.$$

The generic transport identity (10.4) now has strictly positive density and crossing weight for f_*. Thus $\partial_\xi \Psi < 0$, and Proposition 6.1 proves that f_* is strict PF_4.

12.3 Nonreal Fourier zeros

With $\widehat{f}(z) = \int_{\mathbb{R}} f(x)e^{-izx}\,\mathrm{d}x$, Gaussian translation gives

$$\widehat{f_*}(z) = \sqrt{\frac{\pi}{c}}e^{-z^2/(4c)} \sum_{k=-3}^{3} b_k e^{ck^2} e^{-ikz}. \tag{12.8}$$

Put $w = e^{-iz}$ and $u = w + w^{-1}$. The Laurent factor in (12.8) vanishes exactly when

$$R_c(u) = 2e^{9c}u^3 + 10e^{4c}u^2 + (23e^c - 6e^{9c})u + 30 - 20e^{4c} \tag{12.9}$$

vanishes. At $c = 1/64$, directed 192-bit arithmetic gives

$$\mathrm{disc}(R_c) \in -6.0446137251764984704554480068690598 + [-3.14, 3.14]10^{-34},$$

which is strictly negative. Thus the real cubic has a nonreal conjugate pair. For $|w| = 1$, however, $w+w^{-1}$ is real. The corresponding w-roots therefore lie off the unit circle and lift through $w = e^{-iz}$ to zeros with $\operatorname{Im} z \neq 0$. The Gaussian factor is zero-free. This completes the proof of Theorem 1.3.

13 Reproducibility and evidence boundary

The paper has six replay boundaries.

Object	Quick repository verifier	Role
Global q, F_2, strict PF_3	`scripts/verify_pf3_reduction.py`	algebra, tail, and cross-check audit
Global $\mathcal{C}_4 > 0$	`scripts/verify_c4_confluent.py`	algebra, tail, and cross-check audit
Quotient reduction	`scripts/verify_pf4_wronskian_reduction.py`	exact generic algebra
Transport completion	`scripts/verify_pf4_transport_kernel.py`	exact algebra and checks
Origin order-five obstruction	`scripts/verify_pf5_origin.py`	exact rational
Continuous strict-PF_4 separator	`scripts/verify_continuous_pf4_separator.py`	exact and directed

The two theorem-producing compact covers are separate from the quick wrappers:

Cover	Full directed command
q, F_2, 7731 cells	`work/2026-07-12-riemann-kernel-pf34-classification/certify_pf3_curvature.py`
$\mathcal{C}_4$, 8050 cells	`work/2026-07-13-riemann-kernel-pf4-verification/certify_c4.py`

Both use Arb at 192-bit precision and seven retained theta terms. The release manifest records the complete expected stdout hashes. Their identifying prefixes are

$$\texttt{a52a360cdc5e0} \quad (q, F_2), \qquad \texttt{bf2436a454ece2c4} \quad (\mathcal{C}_4).$$

The PF3 quick wrapper separately uses a 256-bit Arb jet cross-check; the C4 quick wrapper uses exact SymPy algebra and mpmath diagnostics. Thus wrapper precision is not reported as base-cover precision.

The compact inequalities are directed Arb calculations at 192-bit precision; the tail uses explicit analytic envelopes and rational coefficient bounds. The origin order-five certificate is independent: it uses only standard-library integer arithmetic with outward rounding on a rational grid, Machin's formula, alternating Taylor sums, three retained theta modes, and one geometric tail bound. The quotient, curvature, primitive, endpoint, density-ratio, and confluent-limit identities are mathematical identities for generic smooth functions. The paper displays these identities; the scripts independently replay them.

A focused independent audit should proceed in this order:

1. verify the analytic theta normalization in (3.1);
2. replay the directed $q, F_2, \mathcal{C}_4$ compact and tail certificates;
3. check the sign $\mathcal{T} \log A = M$ and the two weighted-mean splits in Section 6;
4. expand the determinant and curvature identity in Appendix A;
5. check the endpoint identity in Appendix C;
6. differentiate the density ratio in (10.2);

7. derive the parity factorization and divided-difference limit in Section 11, then replay the rational origin certificate;

8. reconstruct the separator's central determinant; then verify the coefficient signs and Fourier discriminant in Appendix E.

No numerical scan is used as evidence for global nonconfluent positivity of either kernel.

The non-mutating entry point `scripts/replay_paper.py` checks all six registered verifier outputs in quick mode. The option `--full` first reruns both compact covers. Exact package versions are in `requirements-paper.txt`.

14 Declarations, availability, and computational provenance

Author contribution. Joshua Rainstar conceived and directed the project, developed the proof architecture, assembled and audited the computational evidence, and prepared the manuscript.

Acknowledgments and computational assistance. The author acknowledges the use of OpenAI Codex and other disclosed AI systems for symbolic derivation, implementation, drafting, and adversarial audit. The proof records preserve the resulting calculations and their replay boundaries; the author takes responsibility for the mathematical claims.

Conflict of interest. The author declares no competing interests.

Code and data availability. The public repository is `https://github.com/falseywinchnet/zeta`. The theorem-producing evidence integrated before this revision is frozen at MIND epoch `P000069`, Git commit `97bd629c0188`. A submission release must freeze this revised source, its generated coefficient table, replay tool, environment lock, and built PDF under one versioned archive record; the PDF hash belongs in that release manifest rather than recursively inside the PDF.

All numerical inputs are generated by the repository scripts; there is no separate experimental dataset.

A Confluent and curvature algebra

The central-moment Hankel matrix is

$$\begin{pmatrix} 1 & 0 & m_2 & m_3 \\ 0 & m_2 & m_3 & m_4 \\ m_2 & m_3 & m_4 & m_5 \\ m_3 & m_4 & m_5 & m_6 \end{pmatrix}.$$

Taking the Schur complement of the upper-left entry reduces its determinant to

$$\det \begin{pmatrix} m_2 & m_3 & m_4 \\ m_3 & m_4 - m_2^2 & m_5 - m_2 m_3 \\ m_4 & m_5 - m_2 m_3 & m_6 - m_3^2 \end{pmatrix}.$$

Substitution of the moments from Section 4 and ordinary three-by-three expansion gives

$$\begin{aligned}\mathcal{C}_4 = 12c_2^6 + 24c_2^4c_4 - 24c_2^3c_3^2 + 2c_2^3c_6 - 12c_2^2c_3c_5 + 7c_2^2c_4^2\\ + 12c_2c_3^2c_4 + c_2c_4c_6 - c_2c_5^2 - 9c_3^4 - c_3^2c_6 + 2c_3c_4c_5 - c_4^3.\end{aligned}$$

In the curvature coordinate, repeated application of $Q\,\mathrm{d}/\,\mathrm{d}y$ to $c_2 = -Q$ gives

$$\begin{aligned}c_2 &= -Q,\\ c_3 &= -QQ',\\ c_4 &= -Q\{(Q')^2 + QQ''\},\\ c_5 &= -Q\{(Q')^3 + 4QQ'Q'' + Q^2Q'''\},\\ c_6 &= -Q\{(Q')^4 + 11Q(Q')^2Q'' + 4Q^2(Q'')^2 + 7Q^2Q'Q''' + Q^3Q''''\}.\end{aligned}$$

Substitution in the preceding polynomial gives

$$\mathcal{C}_4 = -Q^6\mathcal{P},$$

where

$$\begin{aligned}\mathcal{P} = QQ''Q'''' - Q(Q''')^2 - 2QQ'''' + Q'Q''Q''' - 2Q'Q'''\\ + 3(Q'')^3 - 15(Q'')^2 + 24Q'' - 12.\end{aligned}$$

Now $\kappa = 2 - Q''$, and direct differentiation gives

$$3(\kappa - 1) - \{Q(\log\kappa)'\}' = -\frac{\mathcal{P}}{(Q''-2)^2} = -\frac{\mathcal{P}}{\kappa^2}.$$

Therefore $\mathcal{C}_4 = Q^6\kappa^2 D$, proving (8.2) by an exposed common numerator.

B Tail envelopes and normalized margin

For $x = \pi e^{2t}$, write

$$\Phi(t) = 4\pi x e^{5t/2-x}(1+\rho(x)), \qquad \rho(x) = -\frac{3}{2x} + \sigma(x), \qquad w = \log(1+\rho).$$

The $n \geq 2$ theta terms satisfy, with their first six derivatives, an exponentially small bound absorbed into the following rational paddings for $x \geq \pi e^2$:

$$|w^{(k)}(x)| \leq \frac{C_k}{x^{k+1}},$$

k	1	2	3	4	5	6
C_k	1.61	3.33	10.4	42.8	222	1376

For $D_x = 2x\,\mathrm{d}/\,\mathrm{d}x$, induction gives the Stirling identity

$$D_x^j\psi = 2^j\sum_{k=1}^{j} S(j,k)x^k\psi^{(k)},$$

where $S(j,k)$ are Stirling numbers of the second kind. Since only w contributes to the error in derivatives of order at least two,

$$|\varepsilon_j| \le \frac{E_j}{x}, \qquad E_j = 2^j \sum_{k=1}^{j} S(j,k) C_k.$$

Rounding each sum upward gives

j	2	3	4	5	6
E_j	19.8	176	2082	30770	545900.

For example,

$$E_4 \le 16(C_1 + 7C_2 + 6C_3 + C_4) = 2081.92 < 2082.$$

The logarithmic derivative formulas for w are finite Faa di Bruno sums; the $n = 1$ contribution uses $\rho' \sim 3/(2x^2), \rho'' \sim -3/x^3, \ldots$, while the common theta-tail padding controls σ and all its derivatives. These finite derivative paddings are a directed certificate boundary: the verifier evaluates their Faa di Bruno inequalities with outward-rounded balls. The article records the resulting constants and the analytic monotonicity argument below, but does not claim to print every interval operation.

Substituting $c_j = -2^j x + \delta_j E_j/x$, $|\delta_j| \le 1$, into the thirteen-term polynomial and collecting absolute coefficients gives

$$\begin{aligned} P(x) = 49152x^6 &- \frac{7299072}{5}x^4 - \frac{88018944}{5}x^3 - \frac{756929024}{5}x^2 - \frac{9307871232}{25}x \\ &- \frac{90509073984}{25} - \frac{1807239327472}{125x} - \frac{3528006311808}{125x^2} - \frac{62513985514124}{625x^3} \\ &- \frac{220885105972212}{3125x^4} - \frac{8406623961648}{625x^5} - \frac{11297761792812}{15625x^6}. \end{aligned}$$

Every subtracted term in $P(x)/x^6$ has the form $b_d x^{d-6}$ with $b_d > 0$ and $d < 6$. Therefore

$$\frac{\mathrm{d}}{\mathrm{d}x} \frac{P(x)}{x^6} = \sum_{d<6} (6-d) b_d x^{d-7} > 0.$$

The normalized margin is increasing, so its minimum on $x \ge \pi e^2$ occurs at the left endpoint. Evaluation at the rational lower bound $x > 1157/50$ gives

$$\frac{\mathcal{C}_4}{x^6} \ge \frac{P(x)}{x^6} > 44392.80919.$$

C Endpoint audit of the transport cancellation

This appendix records the algebra suppressed by the phrase "simplifying the endpoint expression." Put

$$z = p + L, \qquad w = z + R, \qquad a = \frac{Q(z) - Q(p)}{L}, \qquad b = \frac{Q(w) - Q(z)}{R}.$$

Direct integration of $2 - Q''$ gives

$$\delta = \frac{L + Q'(p) - a}{L}, \qquad \Lambda = L + R + a - b. \tag{C.1}$$

Define

$$J_y = y^3 - 3yQ(y) + Q(y)Q'(y),$$

$$H_p = 3p^2 - 3Q(p) - 3pQ'(p) + Q'(p)^2 + Q(p)Q''(p),$$

and, for endpoint data,

$$U(c,d) = Q(c) + Q(d) - \frac{Q(d)Q'(d) - Q(c)Q'(c)}{d-c} + \frac{(Q(d)-Q(c))^2}{(d-c)^2}.$$

With $I_L = J_z - J_p$ and $I_R = J_w - J_z$, the identity needed in Lemma 9.1 is exactly

$$\begin{aligned} &\frac{-(J_z - J_p)/L + (J_w - J_z)/R}{\Lambda} + \frac{LH_p - (J_z - J_p)}{L^2\delta} \\ &\quad = \Lambda + Q'(p) - a + \frac{U(p,z) - U(z,w)}{\Lambda} + \frac{U(p,z) - Q(p)(2 - Q''(p))}{L\delta}. \end{aligned}$$

To check it, substitute (C.1), replace z by $p + L$ and w by $p + L + R$, and multiply by the common denominator $L^2 R\delta\Lambda$. The terms containing $Q'(w)$, then $Q'(z)$, then $Q''(p)$ cancel pairwise; the remaining endpoint values and position terms cancel after inserting $a = (Q(z) - Q(p))/L$ and $b = (Q(w) - Q(z))/R$. No differential equation for Q is used, so the identity holds for arbitrary smooth Q.

D Proof provenance

The maintained source map is `paper/PAPER_MAP.md`. The principal repository proof records are:

Claim	Proof record
Global q, F_2, order three	`sources/pf3-global-certificate.md`
Global confluent $\mathcal{C}_4$	`sources/c4-confluent-certificate.md`
Quotient and three-point equivalence	`sources/pf4-wronskian-reduction-certificate.md`
Transport identity and completion	`sources/pf4-transport-kernel-certificate.md`
Origin order-five obstruction	`sources/pf5-origin-certificate.md`
Continuous strict-PF_4 Fourier separator	`sources/continuous-pf4-separator-certificate.md`

The corresponding MIND identities are

$$R142\text{–}R144, \quad R147\text{–}R152, \quad R154\text{–}R156, \quad R153, R164,$$
$$R14, R37, R50, R72, \quad R165\text{–}R170.$$

The replay boundaries are

$$CERT2,\ CERT3,\ CERT5,\ CERT9,\ CERT11,\ CERT10.$$

These identifiers are evidence anchors, not bibliographic citations.

E Exact separator audit

This appendix records the algebraic boundary behind (12.7). Starting from the cumulants $\ell_2, \ldots, \ell_6$, the verifier reconstructs the central moments

$$m_0 = 1, \quad m_1 = 0, \quad m_2 = \ell_2, \quad m_3 = \ell_3,$$

$$m_4 = \ell_4 + 3\ell_2^2, \qquad m_5 = \ell_5 + 10\ell_3\ell_2,$$
$$m_6 = \ell_6 + 15\ell_4\ell_2 + 10\ell_3^2 + 15\ell_2^3,$$

and then forms $\det[m_{i+j}]_{i,j=0}^3$ directly. This independently regenerates the thirteen-term $\mathcal{C}_4$ polynomial; it does not import the expansion used for the Riemann kernel.

The recurrence (12.5) is then applied over $\mathbb{Q}[t]$. Substitution of (12.6) shows that each of the thirteen determinant monomials has common denominator P^{12}. After collecting the numerator H, exact rational arithmetic verifies

$$\deg H = 72, \qquad [t^j]H = [t^{72-j}]H > 0 \quad (0 \le j \le 72),$$

with

$$\min_{0\le j\le 72} [t^j]H = \frac{3}{65536}.$$

For a printable certificate, write

$$2^{54} H(t) = \sum_{j=0}^{72} a_j t^j, \qquad a_j = a_{72-j}.$$

The 37 independent integer coefficients are:

j	a_j	j	a_j
0	824633720832	19	6987741510406573876435979520
1	48712961228800	20	17092065810580803244866465328
2	1429432562155520	21	39268975295924090460974938880
3	27770895893217280	22	84898287500864044148851126266
4	401708065273655296	23	173000552553192254015824625680
5	4613079582269972480	24	332750476261206454685845998080
6	43792294989734840832	25	604872001890033837007447191280
7	353349130016052428800	26	1040322081625845770286033564593
8	2472885380960577236992	27	1694567248719256234936725175440
9	15243207814279730964480	28	2616435044092694236816009466532
10	83763600430517403650816	29	3832176384214979954574258944560
11	414327966630038761815040	30	5327751824843881176247055457873
12	1859451652398640291313152	31	7034632066783174173709968967280
13	7621494792057501544629760	32	8825381090424751586774701723088
14	28689460125397617502291392	33	10523927205327676216804189288720
15	99652010444258917615558400	34	11931512643001413047042570621636
16	320698275507551388688130816	35	12863883307375841645695614141760
17	959582391519228973730931840	36	13190420295966739733605724579768
18	2677774483831421514536291072		

This table is generated by `scripts/generate_separator_coefficients.py`. Its machine-readable companion `generated/separator-coefficients.json` has SHA-256 prefix `87af00802c0929ca`; the complete hash is recorded in the release manifest. Coefficient positivity is a proof on the full domain $t > 0$, not a sampled test. Exact spot reconstructions at $t = 1/3, 1, 4$ independently compare the central determinant with H/P^{12}.

For the Fourier step, the identities

$$w + w^{-1} = u, \quad w^2 + w^{-2} = u^2 - 2, \quad w^3 + w^{-3} = u^3 - 3u$$

give (12.9) directly. The only directed numerical step is the 192-bit Arb enclosure of its discriminant. All remaining checks are symbolic or exact rational arithmetic.

References

[1] A. Belton, D. Guillot, A. Khare, and M. Putinar, *Preservers of totally positive kernels and Polya frequency functions*, arXiv:2110.08206.

[2] A. Khare, *Multiply positive functions, critical exponent phenomena, and the Jain–Karlin–Schoenberg kernel*, arXiv:2008.05121v2.

[3] D. K. Dimitrov and Y. Xu, *Wronskians of Fourier and Laplace transforms*, arXiv:1606.05011.

[4] G. Csordas and R. S. Varga, *Moment inequalities and the Riemann Hypothesis*, Constructive Approximation **4** (1988), 175–198, doi:10.1007/BF02075457.